Newman-Penrose Formalism Calculations

Stephani et al., Equation (17.16)

Metric

$$ds^2 = -\frac{1}{x}\mathrm{d}t \otimes \mathrm{d}t + \frac{by}{ax}\mathrm{d}t \otimes \mathrm{d}\phi + \frac{(\alpha^2 - 2\,\alpha + 1)x}{2\,(ax^2e^y - b^2e^y)}\mathrm{d}x \otimes \mathrm{d}x + \frac{1}{2}\,((\alpha^2e^y - 2\,\alpha e^y + e^y)xy - (\alpha^3(e^y - 1) - 2\,\alpha^2(e^y - 1) + \alpha(e^y - 1))x)e^{(-2\,y)}\mathrm{d}y \otimes \mathrm{d}y + \frac{by}{ax}\mathrm{d}\phi \otimes \mathrm{d}t - \frac{(a\alpha x^2(e^y - 1) - ax^2ye^y + b^2y^2e^y)e^{(-y)}}{a^2x}\mathrm{d}\phi \otimes \mathrm{d}\phi$$

Coordinates

$(x^\mu) = (t, x, y, \phi)$

Null Tetrad

Covariant components

$$l = (\frac{\sqrt{2}}{2\,\sqrt{x}})dt + (\frac{1}{2}\,\sqrt{2}\sqrt{\frac{1}{2}}\sqrt{\frac{(\alpha - 1)^2x}{ax^2 - b^2}}e^{(-\frac{1}{2}\,y)})dx + (0)dy + (-\frac{\sqrt{2}by}{2\,a\sqrt{x}})d\phi \tag{1}$$

$$n = (\frac{\sqrt{2}}{2\,\sqrt{x}})dt + (-\frac{1}{2}\,\sqrt{2}\sqrt{\frac{1}{2}}\sqrt{\frac{(\alpha - 1)^2x}{ax^2 - b^2}}e^{(-\frac{1}{2}\,y)})dx + (0)dy + (-\frac{\sqrt{2}by}{2\,a\sqrt{x}})d\phi \tag{2}$$

$$m = (0)dt + (0)dx + (\frac{1}{2}\,\sqrt{2}\sqrt{\frac{1}{2}}\sqrt{(\alpha(e^{(-y)} - 1) + y)(\alpha - 1)^2x}e^{(-\frac{1}{2}\,y)})dy + (\frac{1}{2}i\,\sqrt{2}\sqrt{\frac{(\alpha(e^{(-y)} - 1) + y)x}{a}})d\phi \tag{3}$$

$$\overline{m} = (0)dt + (0)dx + (\frac{1}{2}\,\sqrt{2}\sqrt{\frac{1}{2}}\sqrt{(\alpha(e^{(-y)} - 1) + y)(\alpha - 1)^2x}e^{(-\frac{1}{2}\,y)})dy + (-\frac{1}{2}i\,\sqrt{2}\sqrt{\frac{(\alpha(e^{(-y)} - 1) + y)x}{a}})d\phi \tag{4}$$

Contravariant components

$$l = (-\frac{1}{2}\sqrt{2}\sqrt{x})dt + (\frac{ax^2|\alpha-1|e^{(\frac{1}{2}y)} - b^2|\alpha-1|e^{(\frac{1}{2}y)}}{\sqrt{ax^2-b^2}(\alpha^2-2\alpha+1)\sqrt{x}})dx + (0)dy + (0)d\phi \tag{5}$$

$$n = (-\frac{1}{2}\sqrt{2}\sqrt{x})dt + (-\frac{ax^2|\alpha-1|e^{(\frac{1}{2}y)} - b^2|\alpha-1|e^{(\frac{1}{2}y)}}{\sqrt{ax^2-b^2}(\alpha^2-2\alpha+1)\sqrt{x}})dx + (0)dy + (0)d\phi \tag{6}$$

$$m = (\frac{i\sqrt{-\alpha(e^y-1)+ye^y}bye^{(\frac{1}{2}y)}}{(\sqrt{2}ye^y-(\sqrt{2}e^y-\sqrt{2})\alpha)\sqrt{a}\sqrt{x}})dt + (0)dx + (\frac{e^y}{\sqrt{-\alpha(e^y-1)+ye^y}\sqrt{x}|\alpha-1|})dy + (\frac{i\sqrt{-\alpha(e^y-1)+ye^y}\sqrt{a}e^{(\frac{1}{2}y)}}{(\sqrt{2}ye^y-(\sqrt{2}e^y-\sqrt{2})\alpha)\sqrt{x}})d\phi \tag{7}$$

$$\overline{m} = (-\frac{i\sqrt{-\alpha(e^y-1)+ye^y}bye^{(\frac{1}{2}y)}}{(\sqrt{2}ye^y-(\sqrt{2}e^y-\sqrt{2})\alpha)\sqrt{a}\sqrt{x}})dt + (0)dx + (\frac{e^y}{\sqrt{-\alpha(e^y-1)+ye^y}\sqrt{x}|\alpha-1|})dy + (-\frac{i\sqrt{-\alpha(e^y-1)+ye^y}\sqrt{a}e^{(\frac{1}{2}y)}}{(\sqrt{2}ye^y-(\sqrt{2}e^y-\sqrt{2})\alpha)\sqrt{x}})d\phi \tag{8}$$

Directional Derivatives

$$D = -\sqrt{2}e^{(-\frac{1}{2}\lambda(u,v))}\,\partial_v$$

$$\Delta = -\sqrt{2}e^{(-\frac{1}{2}\lambda(u,v))}\,\partial_u$$

$$\delta = -\frac{\sqrt{2}e^{(\frac{1}{2}\mu(u,v))}}{u-v}\,\partial_\varphi + \frac{i\sqrt{2}e^{(-\frac{1}{2}\mu(u,v))}}{u+v}\,\partial_\chi$$

$$\overline{\delta} = -\frac{\sqrt{2}e^{(\frac{1}{2}\mu(u,v))}}{u-v}\,\partial_\varphi - \frac{i\sqrt{2}e^{(-\frac{1}{2}\mu(u,v))}}{u+v}\,\partial_\chi$$

Spin Coefficients

$$\rho = -\frac{\sqrt{ax^2-b^2}\alpha e^{(\frac{3}{2}\,y)}}{2\,(\alpha x|\alpha-1|e^y - xy|\alpha-1|e^y - \alpha x|\alpha-1|)\sqrt{x}} + \frac{\sqrt{ax^2-b^2}y e^{(\frac{3}{2}\,y)}}{2\,(\alpha x|\alpha-1|e^y - xy|\alpha-1|e^y - \alpha x|\alpha-1|)\sqrt{x}} + \frac{\sqrt{ax^2-b^2}\alpha e^{(\frac{1}{2}\,y)}}{2\,(\alpha x|\alpha-1|e^y - xy|\alpha-1|e^y - \alpha x|\alpha-1|)\sqrt{x}} - \frac{i\,b e^{(\frac{3}{2}\,y)}}{2\,(\alpha x|\alpha-1|e^y - xy|\alpha-1|e^y - \alpha x|\alpha-1|)\sqrt{a}\sqrt{x}} \tag{9}$$

$$\sigma = 0 \tag{10}$$

$$\kappa = -\frac{|\alpha-1|e^y}{4\,(\alpha^2-2\,\alpha+1)\sqrt{-\alpha e^y + y e^y + \alpha}\sqrt{x}} \tag{11}$$

$$\mu = -\frac{\sqrt{ax^2-b^2}\alpha e^{(\frac{3}{2}\,y)}}{2\,(\alpha x|\alpha-1|e^y - xy|\alpha-1|e^y - \alpha x|\alpha-1|)\sqrt{x}} + \frac{\sqrt{ax^2-b^2}y e^{(\frac{3}{2}\,y)}}{2\,(\alpha x|\alpha-1|e^y - xy|\alpha-1|e^y - \alpha x|\alpha-1|)\sqrt{x}} + \frac{\sqrt{ax^2-b^2}\alpha e^{(\frac{1}{2}\,y)}}{2\,(\alpha x|\alpha-1|e^y - xy|\alpha-1|e^y - \alpha x|\alpha-1|)\sqrt{x}} - \frac{i\,b e^{(\frac{3}{2}\,y)}}{2\,(\alpha x|\alpha-1|e^y - xy|\alpha-1|e^y - \alpha x|\alpha-1|)\sqrt{a}\sqrt{x}} \tag{12}$$

$$\lambda = 0 \tag{13}$$

$$\nu = \frac{|\alpha-1|e^y}{4\,(\alpha^2-2\,\alpha+1)\sqrt{-\alpha e^y + y e^y + \alpha}\sqrt{x}} \tag{14}$$

$$\alpha = -\frac{\alpha e^y}{4\,(\alpha|\alpha-1|e^y - y|\alpha-1|e^y - \alpha|\alpha-1|)\sqrt{-\alpha e^y + y e^y + \alpha}\sqrt{x}} + \frac{e^{(2\,y)}}{4\,(\alpha|\alpha-1|e^y - y|\alpha-1|e^y - \alpha|\alpha-1|)\sqrt{-\alpha e^y + y e^y + \alpha}\sqrt{x}} \tag{15}$$

$$\beta = \frac{\alpha e^y}{4\,(\alpha|\alpha-1|e^y - y|\alpha-1|e^y - \alpha|\alpha-1|)\sqrt{-\alpha e^y + y e^y + \alpha}\sqrt{x}} - \frac{e^{(2\,y)}}{4\,(\alpha|\alpha-1|e^y - y|\alpha-1|e^y - \alpha|\alpha-1|)\sqrt{-\alpha e^y + y e^y + \alpha}\sqrt{x}} \tag{16}$$

$$\tau = \frac{|\alpha - 1| e^{y}}{4\,(\alpha^{2} - 2\,\alpha + 1)\sqrt{-\alpha e^{y} + y e^{y} + \alpha}\sqrt{x}} \tag{17}$$

$$\pi = -\frac{|\alpha - 1| e^{y}}{4\,(\alpha^{2} - 2\,\alpha + 1)\sqrt{-\alpha e^{y} + y e^{y} + \alpha}\sqrt{x}} \tag{18}$$

$$\begin{aligned}\epsilon = &-\frac{a\alpha x^{\frac{3}{2}} e^{(\frac{3}{2}\,y)}}{4\,(\alpha x|\alpha - 1|e^{y} - xy|\alpha - 1|e^{y} - \alpha x|\alpha - 1|)\sqrt{a x^{2} - b^{2}}} + \frac{a x^{\frac{3}{2}} y e^{(\frac{3}{2}\,y)}}{4\,(\alpha x|\alpha - 1|e^{y} - xy|\alpha - 1|e^{y} - \alpha x|\alpha - 1|)\sqrt{a x^{2} - b^{2}}} + \frac{a\alpha x^{\frac{3}{2}} e^{(\frac{1}{2}\,y)}}{4\,(\alpha x|\alpha - 1|e^{y} - xy|\alpha - 1|e^{y} - \alpha x|\alpha - 1|)\sqrt{a x^{2} - b^{2}}} \\ &+ \frac{\alpha b^{2} e^{(\frac{3}{2}\,y)}}{4\,(\alpha x|\alpha - 1|e^{y} - xy|\alpha - 1|e^{y} - \alpha x|\alpha - 1|)\sqrt{a x^{2} - b^{2}}\sqrt{x}} - \frac{b^{2} y e^{(\frac{3}{2}\,y)}}{4\,(\alpha x|\alpha - 1|e^{y} - xy|\alpha - 1|e^{y} - \alpha x|\alpha - 1|)\sqrt{a x^{2} - b^{2}}\sqrt{x}} \\ &- \frac{\alpha b^{2} e^{(\frac{1}{2}\,y)}}{4\,(\alpha x|\alpha - 1|e^{y} - xy|\alpha - 1|e^{y} - \alpha x|\alpha - 1|)\sqrt{a x^{2} - b^{2}}\sqrt{x}} - \frac{i\,b e^{(\frac{3}{2}\,y)}}{4\,(\alpha x|\alpha - 1|e^{y} - xy|\alpha - 1|e^{y} - \alpha x|\alpha - 1|)\sqrt{a}\sqrt{x}}\end{aligned} \tag{19}$$

$$\begin{aligned}\gamma = &-\frac{a\alpha x^{\frac{3}{2}} e^{(\frac{3}{2}\,y)}}{4\,(\alpha x|\alpha - 1|e^{y} - xy|\alpha - 1|e^{y} - \alpha x|\alpha - 1|)\sqrt{a x^{2} - b^{2}}} + \frac{a x^{\frac{3}{2}} y e^{(\frac{3}{2}\,y)}}{4\,(\alpha x|\alpha - 1|e^{y} - xy|\alpha - 1|e^{y} - \alpha x|\alpha - 1|)\sqrt{a x^{2} - b^{2}}} + \frac{a\alpha x^{\frac{3}{2}} e^{(\frac{1}{2}\,y)}}{4\,(\alpha x|\alpha - 1|e^{y} - xy|\alpha - 1|e^{y} - \alpha x|\alpha - 1|)\sqrt{a x^{2} - b^{2}}} \\ &+ \frac{\alpha b^{2} e^{(\frac{3}{2}\,y)}}{4\,(\alpha x|\alpha - 1|e^{y} - xy|\alpha - 1|e^{y} - \alpha x|\alpha - 1|)\sqrt{a x^{2} - b^{2}}\sqrt{x}} - \frac{b^{2} y e^{(\frac{3}{2}\,y)}}{4\,(\alpha x|\alpha - 1|e^{y} - xy|\alpha - 1|e^{y} - \alpha x|\alpha - 1|)\sqrt{a x^{2} - b^{2}}\sqrt{x}} \\ &- \frac{\alpha b^{2} e^{(\frac{1}{2}\,y)}}{4\,(\alpha x|\alpha - 1|e^{y} - xy|\alpha - 1|e^{y} - \alpha x|\alpha - 1|)\sqrt{a x^{2} - b^{2}}\sqrt{x}} - \frac{i\,b e^{(\frac{3}{2}\,y)}}{4\,(\alpha x|\alpha - 1|e^{y} - xy|\alpha - 1|e^{y} - \alpha x|\alpha - 1|)\sqrt{a}\sqrt{x}}\end{aligned} \tag{20}$$

Ricci Tensor Components

$$\Phi_{00} = -\frac{a^{2}\alpha^{2} x^{2} e^{(3y)}}{4\,D_{1}} + \frac{a^{2}\alpha x^{2} y e^{(3y)}}{2\,D_{1}} - \frac{a^{2} x^{2} y^{2} e^{(3y)}}{4\,D_{1}} + \frac{a^{2}\alpha^{2} x^{2} e^{(2y)}}{2\,D_{1}} - \frac{a^{2}\alpha x^{2} y e^{(2y)}}{2\,D_{1}} - \frac{a^{2}\alpha^{2} x^{2} e^{y}}{4\,D_{1}} - \frac{a\alpha^{2} b^{2} e^{(3y)}}{4\,D_{1}} + \frac{a\alpha b^{2} y e^{(3y)}}{2\,D_{1}} - \frac{a b^{2} y^{2} e^{(3y)}}{4\,D_{1}} + \frac{a\alpha^{2} b^{2} e^{(2y)}}{2\,D_{1}} - \frac{a\alpha b^{2} y e^{(2y)}}{2\,D_{1}} - \frac{a\alpha^{2} b^{2} e^{y}}{4\,D_{1}} + \frac{b^{2} e^{(3y)}}{4\,D_{1}} \tag{21}$$

$$\begin{aligned}D_{1} = &\,a\alpha^{4} x^{3} e^{(2y)} - 2\,a\alpha^{3} x^{3} y e^{(2y)} + a\alpha^{2} x^{3} y^{2} e^{(2y)} - 2\,a\alpha^{4} x^{3} e^{y} + 2\,a\alpha^{3} x^{3} y e^{y} + a\alpha^{4} x^{3} - 2\,a\alpha^{3} x^{3} e^{(2y)} + 4\,a\alpha^{2} x^{3} y e^{(2y)} - 2\,a\alpha x^{3} y^{2} e^{(2y)} \\ &+ 4\,a\alpha^{3} x^{3} e^{y} - 4\,a\alpha^{2} x^{3} y e^{y} - 2\,a\alpha^{3} x^{3} + a\alpha^{2} x^{3} e^{(2y)} - 2\,a\alpha x^{3} y e^{(2y)} + a x^{3} y^{2} e^{(2y)} - 2\,a\alpha^{2} x^{3} e^{y} + 2\,a\alpha x^{3} y e^{y} + a\alpha^{2} x^{3}\end{aligned}$$

$$\Phi_{01} = -\frac{i\sqrt{-\alpha e^{y} + y e^{y} + \alpha}\,\alpha b\, e^{(\frac{5}{2}y)}}{4\,D_2\sqrt{a}} + \frac{i\sqrt{-\alpha e^{y} + y e^{y} + \alpha}\,b\, e^{(\frac{7}{2}y)}}{4\,D_2\sqrt{a}} \tag{22}$$

$$\begin{aligned}D_2 = {} & \alpha^5 x^2 e^{(3y)} - 3\,\alpha^4 x^2 y e^{(3y)} + 3\,\alpha^3 x^2 y^2 e^{(3y)} - \alpha^2 x^2 y^3 e^{(3y)} - 3\,\alpha^5 x^2 e^{(2y)} + 6\,\alpha^4 x^2 y e^{(2y)} - 3\,\alpha^3 x^2 y^2 e^{(2y)} + 3\,\alpha^5 x^2 e^{y} - 3\,\alpha^4 x^2 y e^{y} - \alpha^5 x^2 \\ & - 2\,\alpha^4 x^2 e^{(3y)} + 6\,\alpha^3 x^2 y e^{(3y)} - 6\,\alpha^2 x^2 y^2 e^{(3y)} + 2\,\alpha x^2 y^3 e^{(3y)} + 6\,\alpha^4 x^2 e^{(2y)} - 12\,\alpha^3 x^2 y e^{(2y)} + 6\,\alpha^2 x^2 y^2 e^{(2y)} - 6\,\alpha^4 x^2 e^{y} + 6\,\alpha^3 x^2 y e^{y} \\ & + 2\,\alpha^4 x^2 + \alpha^3 x^2 e^{(3y)} - 3\,\alpha^2 x^2 y e^{(3y)} + 3\,\alpha x^2 y^2 e^{(3y)} - x^2 y^3 e^{(3y)} - 3\,\alpha^3 x^2 e^{(2y)} + 6\,\alpha^2 x^2 y e^{(2y)} - 3\,\alpha x^2 y^2 e^{(2y)} + 3\,\alpha^3 x^2 e^{y} - 3\,\alpha^2 x^2 y e^{y} - \alpha^3 x^2\end{aligned}$$

$$\begin{aligned}\Phi_{02} & \\ = {} & \frac{\alpha e^{(2\,y)}}{4\,(\alpha^4 x e^{(2\,y)} - 2\,\alpha^3 x y e^{(2\,y)} + \alpha^2 x y^2 e^{(2\,y)} - 2\,\alpha^4 x e^{y} + 2\,\alpha^3 x y e^{y} + \alpha^4 x - 2\,\alpha^3 x e^{(2\,y)} + 4\,\alpha^2 x y e^{(2\,y)} - 2\,\alpha x y^2 e^{(2\,y)} + 4\,\alpha^3 x e^{y} - 4\,\alpha^2 x y e^{y} - 2\,\alpha^3 x + \alpha^2 x e^{(2\,y)} - 2\,\alpha x y e^{(2\,y)} + x y^2 e^{(2\,y)} - 2\,\alpha^2 x e^{y} + 2\,\alpha x y e^{y} + \alpha^2 x)} \\ & - \frac{e^{(3\,y)}}{4\,(\alpha^4 x e^{(2\,y)} - 2\,\alpha^3 x y e^{(2\,y)} + \alpha^2 x y^2 e^{(2\,y)} - 2\,\alpha^4 x e^{y} + 2\,\alpha^3 x y e^{y} + \alpha^4 x - 2\,\alpha^3 x e^{(2\,y)} + 4\,\alpha^2 x y e^{(2\,y)} - 2\,\alpha x y^2 e^{(2\,y)} + 4\,\alpha^3 x e^{y} - 4\,\alpha^2 x y e^{y} - 2\,\alpha^3 x + \alpha^2 x e^{(2\,y)} - 2\,\alpha x y e^{(2\,y)} + x y^2 e^{(2\,y)} - 2\,\alpha^2 x e^{y} + 2\,\alpha x y e^{y} + \alpha^2 x)}\end{aligned} \tag{23}$$

$$\Phi_{10} = \frac{i\sqrt{-\alpha e^{y} + y e^{y} + \alpha}\,\alpha b\, e^{(\frac{5}{2}y)}}{4\,D_2\sqrt{a}} - \frac{i\sqrt{-\alpha e^{y} + y e^{y} + \alpha}\,b\, e^{(\frac{7}{2}y)}}{4\,D_2\sqrt{a}} \tag{24}$$

$$\begin{aligned}\Phi_{11} = {} & \frac{a^2\alpha^3 x^2 e^{(4y)}}{8\,D_3} - \frac{3\,a^2\alpha^2 x^2 y e^{(4y)}}{8\,D_3} + \frac{3\,a^2\alpha x^2 y^2 e^{(4y)}}{8\,D_3} - \frac{a^2 x^2 y^3 e^{(4y)}}{8\,D_3} - \frac{3\,a^2\alpha^3 x^2 e^{(3y)}}{8\,D_3} + \frac{3\,a^2\alpha^2 x^2 y e^{(3y)}}{4\,D_3} - \frac{3\,a^2\alpha x^2 y^2 e^{(3y)}}{8\,D_3} + \frac{3\,a^2\alpha^3 x^2 e^{(2y)}}{8\,D_3} - \frac{3\,a^2\alpha^2 x^2 y e^{(2y)}}{8\,D_3} \\ & - \frac{a^2\alpha^3 x^2 e^{y}}{8\,D_3} - \frac{3\,a\alpha^3 b^2 e^{(4y)}}{8\,D_3} + \frac{9\,a\alpha^2 b^2 y e^{(4y)}}{8\,D_3} - \frac{9\,a\alpha b^2 y^2 e^{(4y)}}{8\,D_3} + \frac{3\,a b^2 y^3 e^{(4y)}}{8\,D_3} + \frac{9\,a\alpha^3 b^2 e^{(3y)}}{8\,D_3} - \frac{9\,a\alpha^2 b^2 y e^{(3y)}}{4\,D_3} + \frac{9\,a\alpha b^2 y^2 e^{(3y)}}{8\,D_3} - \frac{9\,a\alpha^3 b^2 e^{(2y)}}{8\,D_3} + \frac{9\,a\alpha^2 b^2 y e^{(2y)}}{8\,D_3} \\ & + \frac{3\,a\alpha^3 b^2 e^{y}}{8\,D_3} - \frac{a\alpha^2 x^2 e^{(3y)}}{8\,D_3} + \frac{a\alpha x^2 y e^{(3y)}}{8\,D_3} - \frac{a\alpha^2 x^2 e^{(2y)}}{8\,D_3} - \frac{a\alpha x^2 e^{(4y)}}{8\,D_3} + \frac{a x^2 y e^{(4y)}}{8\,D_3} + \frac{5\,a\alpha x^2 e^{(3y)}}{8\,D_3} + \frac{3\,\alpha b^2 e^{(4y)}}{8\,D_3} - \frac{a x^2 e^{(4y)}}{4\,D_3} - \frac{3\,b^2 y e^{(4y)}}{8\,D_3} - \frac{3\,\alpha b^2 e^{(3y)}}{8\,D_3}\end{aligned} \tag{25}$$

$$\begin{aligned}D_3 = {} & a\alpha^5 x^3 e^{(3y)} - 3\,a\alpha^4 x^3 y e^{(3y)} + 3\,a\alpha^3 x^3 y^2 e^{(3y)} - a\alpha^2 x^3 y^3 e^{(3y)} - 3\,a\alpha^5 x^3 e^{(2y)} + 6\,a\alpha^4 x^3 y e^{(2y)} - 3\,a\alpha^3 x^3 y^2 e^{(2y)} + 3\,a\alpha^5 x^3 e^{y} - 3\,a\alpha^4 x^3 y e^{y} - a\alpha^5 x^3 \\ & - 2\,a\alpha^4 x^3 e^{(3y)} + 6\,a\alpha^3 x^3 y e^{(3y)} - 6\,a\alpha^2 x^3 y^2 e^{(3y)} + 2\,a\alpha x^3 y^3 e^{(3y)} + 6\,a\alpha^4 x^3 e^{(2y)} - 12\,a\alpha^3 x^3 y e^{(2y)} + 6\,a\alpha^2 x^3 y^2 e^{(2y)} - 6\,a\alpha^4 x^3 e^{y} + 6\,a\alpha^3 x^3 y e^{y} + 2\,a\alpha^4 x^3 \\ & + a\alpha^3 x^3 e^{(3y)} - 3\,a\alpha^2 x^3 y e^{(3y)} + 3\,a\alpha x^3 y^2 e^{(3y)} - a x^3 y^3 e^{(3y)} - 3\,a\alpha^3 x^3 e^{(2y)} + 6\,a\alpha^2 x^3 y e^{(2y)} - 3\,a\alpha x^3 y^2 e^{(2y)} + 3\,a\alpha^3 x^3 e^{y} - 3\,a\alpha^2 x^3 y e^{y} - a\alpha^3 x^3\end{aligned}$$

$$\Phi_{12} = -\frac{i\sqrt{-\alpha e^{y} + y e^{y} + \alpha}\,\alpha b\, e^{(\frac{5}{2}y)}}{4\,D_2\sqrt{a}} + \frac{i\sqrt{-\alpha e^{y} + y e^{y} + \alpha}\,b\, e^{(\frac{7}{2}y)}}{4\,D_2\sqrt{a}} \tag{26}$$

$$\Phi_{20} = \frac{\alpha e^{(2\,y)}}{4\,(\alpha^4 x e^{(2\,y)} - 2\,\alpha^3 x y e^{(2\,y)} + \alpha^2 x y^2 e^{(2\,y)} - 2\,\alpha^4 x e^y + 2\,\alpha^3 x y e^y + \alpha^4 x - 2\,\alpha^3 x e^{(2\,y)} + 4\,\alpha^2 x y e^{(2\,y)} - 2\,\alpha x y^2 e^{(2\,y)} + 4\,\alpha^3 x e^y - 4\,\alpha^2 x y e^y - 2\,\alpha^3 x + \alpha^2 x e^{(2\,y)} - 2\,\alpha x y e^{(2\,y)} + x y^2 e^{(2\,y)} - 2\,\alpha^2 x e^y + 2\,\alpha x y e^y + \alpha^2 x)} - \frac{e^{(3\,y)}}{4\,(\alpha^4 x e^{(2\,y)} - 2\,\alpha^3 x y e^{(2\,y)} + \alpha^2 x y^2 e^{(2\,y)} - 2\,\alpha^4 x e^y + 2\,\alpha^3 x y e^y + \alpha^4 x - 2\,\alpha^3 x e^{(2\,y)} + 4\,\alpha^2 x y e^{(2\,y)} - 2\,\alpha x y^2 e^{(2\,y)} + 4\,\alpha^3 x e^y - 4\,\alpha^2 x y e^y - 2\,\alpha^3 x + \alpha^2 x e^{(2\,y)} - 2\,\alpha x y e^{(2\,y)} + x y^2 e^{(2\,y)} - 2\,\alpha^2 x e^y + 2\,\alpha x y e^y + \alpha^2 x)} \tag{27}$$

$$\Phi_{21} = \frac{i\,\sqrt{-\alpha e^y + y e^y + \alpha}\,\alpha b\,e^{(\frac{5}{2}y)}}{4\,D_2\,\sqrt{a}} - \frac{i\,\sqrt{-\alpha e^y + y e^y + \alpha}\,b\,e^{(\frac{7}{2}y)}}{4\,D_2\,\sqrt{a}} \tag{28}$$

$$\Phi_{22} = -\frac{a^2\alpha^2 x^2 e^{(3y)}}{4\,D_1} + \frac{a^2\alpha x^2 y e^{(3y)}}{2\,D_1} - \frac{a^2 x^2 y^2 e^{(3y)}}{4\,D_1} + \frac{a^2\alpha^2 x^2 e^{(2y)}}{2\,D_1} - \frac{a^2\alpha x^2 y e^{(2y)}}{2\,D_1} - \frac{a^2\alpha^2 x^2 e^y}{4\,D_1} - \frac{a\alpha^2 b^2 e^{(3y)}}{4\,D_1} + \frac{a\alpha b^2 y e^{(3y)}}{2\,D_1} - \frac{a b^2 y^2 e^{(3y)}}{4\,D_1} + \frac{a\alpha^2 b^2 e^{(2y)}}{2\,D_1} - \frac{a\alpha b^2 y e^{(2y)}}{2\,D_1} - \frac{a\alpha^2 b^2 e^y}{4\,D_1} + \frac{b^2 e^{(3y)}}{4\,D_1} \tag{29}$$

$$\begin{aligned}\Lambda = &-\frac{a^2\alpha^3 x^2 e^{(4y)}}{24\,D_3} + \frac{a^2\alpha^2 x^2 y e^{(4y)}}{8\,D_3} - \frac{a^2\alpha x^2 y^2 e^{(4y)}}{8\,D_3} + \frac{a^2 x^2 y^3 e^{(4y)}}{24\,D_3} + \frac{a^2\alpha^3 x^2 e^{(3y)}}{8\,D_3} - \frac{a^2\alpha^2 x^2 y e^{(3y)}}{4\,D_3} + \frac{a^2\alpha x^2 y^2 e^{(3y)}}{8\,D_3} - \frac{a^2\alpha^3 x^2 e^{(2y)}}{8\,D_3} + \frac{a^2\alpha^2 x^2 y e^{(2y)}}{8\,D_3} \\ &+ \frac{a^2\alpha^3 x^2 e^y}{24\,D_3} - \frac{a\alpha^3 b^2 e^{(4y)}}{24\,D_3} + \frac{a\alpha^2 b^2 y e^{(4y)}}{8\,D_3} - \frac{a\alpha b^2 y^2 e^{(4y)}}{8\,D_3} + \frac{a b^2 y^3 e^{(4y)}}{24\,D_3} + \frac{a\alpha^3 b^2 e^{(3y)}}{8\,D_3} - \frac{a\alpha^2 b^2 y e^{(3y)}}{4\,D_3} + \frac{a\alpha b^2 y^2 e^{(3y)}}{8\,D_3} - \frac{a\alpha^3 b^2 e^{(2y)}}{8\,D_3} + \frac{a\alpha^2 b^2 y e^{(2y)}}{8\,D_3} \\ &+ \frac{a\alpha^3 b^2 e^y}{24\,D_3} - \frac{a\alpha^2 x^2 e^{(3y)}}{24\,D_3} + \frac{a\alpha x^2 y e^{(3y)}}{24\,D_3} - \frac{a\alpha^2 x^2 e^{(2y)}}{24\,D_3} - \frac{a\alpha x^2 e^{(4y)}}{24\,D_3} + \frac{a x^2 y e^{(4y)}}{24\,D_3} + \frac{5\,a\alpha x^2 e^{(3y)}}{24\,D_3} + \frac{\alpha b^2 e^{(4y)}}{24\,D_3} - \frac{a x^2 e^{(4y)}}{12\,D_3} - \frac{b^2 y e^{(4y)}}{24\,D_3} - \frac{\alpha b^2 e^{(3y)}}{24\,D_3}\end{aligned} \tag{30}$$

Weyl Tensor Components

$$\Psi_0 = \frac{\alpha e^{(2\,y)}}{4\,(\alpha^4 x e^{(2\,y)} - 2\,\alpha^3 x y e^{(2\,y)} + \alpha^2 x y^2 e^{(2\,y)} - 2\,\alpha^4 x e^y + 2\,\alpha^3 x y e^y + \alpha^4 x - 2\,\alpha^3 x e^{(2\,y)} + 4\,\alpha^2 x y e^{(2\,y)} - 2\,\alpha x y^2 e^{(2\,y)} + 4\,\alpha^3 x e^y - 4\,\alpha^2 x y e^y - 2\,\alpha^3 x + \alpha^2 x e^{(2\,y)} - 2\,\alpha x y e^{(2\,y)} + x y^2 e^{(2\,y)} - 2\,\alpha^2 x e^y + 2\,\alpha x y e^y + \alpha^2 x)} - \frac{e^{(3\,y)}}{4\,(\alpha^4 x e^{(2\,y)} - 2\,\alpha^3 x y e^{(2\,y)} + \alpha^2 x y^2 e^{(2\,y)} - 2\,\alpha^4 x e^y + 2\,\alpha^3 x y e^y + \alpha^4 x - 2\,\alpha^3 x e^{(2\,y)} + 4\,\alpha^2 x y e^{(2\,y)} - 2\,\alpha x y^2 e^{(2\,y)} + 4\,\alpha^3 x e^y - 4\,\alpha^2 x y e^y - 2\,\alpha^3 x + \alpha^2 x e^{(2\,y)} - 2\,\alpha x y e^{(2\,y)} + x y^2 e^{(2\,y)} - 2\,\alpha^2 x e^y + 2\,\alpha x y e^y + \alpha^2 x)} \tag{31}$$

$$\begin{aligned}\Psi_1 = &-\frac{\sqrt{a x^2 - b^2}\,\sqrt{-\alpha e^y + y e^y + \alpha}\,\alpha^2 e^{(\frac{7}{2}y)}}{4\,D_2} + \frac{\sqrt{a x^2 - b^2}\,\sqrt{-\alpha e^y + y e^y + \alpha}\,\alpha y e^{(\frac{7}{2}y)}}{2\,D_2} - \frac{\sqrt{a x^2 - b^2}\,\sqrt{-\alpha e^y + y e^y + \alpha}\,y^2 e^{(\frac{7}{2}y)}}{4\,D_2} \\ &+ \frac{\sqrt{a x^2 - b^2}\,\sqrt{-\alpha e^y + y e^y + \alpha}\,\alpha^2 e^{(\frac{5}{2}y)}}{2\,D_2} - \frac{\sqrt{a x^2 - b^2}\,\sqrt{-\alpha e^y + y e^y + \alpha}\,\alpha y e^{(\frac{5}{2}y)}}{2\,D_2} - \frac{\sqrt{a x^2 - b^2}\,\sqrt{-\alpha e^y + y e^y + \alpha}\,\alpha^2 e^{(\frac{3}{2}y)}}{4\,D_2} \\ &- \frac{i\,\sqrt{-\alpha e^y + y e^y + \alpha}\,\alpha b\,e^{(\frac{7}{2}y)}}{4\,D_2\,\sqrt{a}} + \frac{i\,\sqrt{-\alpha e^y + y e^y + \alpha}\,b y\,e^{(\frac{7}{2}y)}}{4\,D_2\,\sqrt{a}} + \frac{i\,\sqrt{-\alpha e^y + y e^y + \alpha}\,\alpha b\,e^{(\frac{5}{2}y)}}{2\,D_2\,\sqrt{a}} - \frac{i\,\sqrt{-\alpha e^y + y e^y + \alpha}\,b\,e^{(\frac{7}{2}y)}}{4\,D_2\,\sqrt{a}}\end{aligned} \tag{32}$$

$$\Psi_2 = \frac{a^2\alpha^3x^2e^{(4y)}}{3\,D_3} - \frac{a^2\alpha^2x^2ye^{(4y)}}{D_3} + \frac{a^2\alpha x^2y^2e^{(4y)}}{D_3} - \frac{a^2x^2y^3e^{(4y)}}{3\,D_3} - \frac{a^2\alpha^3x^2e^{(3y)}}{D_3} + \frac{2\,a^2\alpha^2x^2ye^{(3y)}}{D_3} - \frac{a^2\alpha x^2y^2e^{(3y)}}{D_3} + \frac{a^2\alpha^3x^2e^{(2y)}}{D_3} - \frac{a^2\alpha^2x^2ye^{(2y)}}{D_3} - \frac{a^2\alpha^3x^2e^{y}}{3\,D_3}$$
$$- \frac{2\,a\alpha^3b^2e^{(4y)}}{3\,D_3} + \frac{2\,a\alpha^2b^2ye^{(4y)}}{D_3} - \frac{2\,a\alpha b^2y^2e^{(4y)}}{D_3} + \frac{2\,ab^2y^3e^{(4y)}}{3\,D_3} + \frac{2\,a\alpha^3b^2e^{(3y)}}{D_3} - \frac{4\,a\alpha^2b^2ye^{(3y)}}{D_3} + \frac{2\,a\alpha b^2y^2e^{(3y)}}{D_3} - \frac{2\,a\alpha^3b^2e^{(2y)}}{D_3} + \frac{2\,a\alpha^2b^2ye^{(2y)}}{D_3} + \frac{2\,a\alpha^3b^2e^{y}}{3\,D_3}$$
$$+ \frac{a\alpha^2x^2e^{(3y)}}{12\,D_3} - \frac{a\alpha x^2ye^{(3y)}}{12\,D_3} + \frac{a\alpha^2x^2e^{(2y)}}{12\,D_3} + \frac{i\sqrt{ax^2-b^2}\sqrt{a}\,\alpha^2b\,e^{(4y)}}{D_3} + \frac{a\alpha x^2e^{(4y)}}{12\,D_3} - \frac{2i\sqrt{ax^2-b^2}\sqrt{a}\,\alpha by\,e^{(4y)}}{D_3} - \frac{ax^2ye^{(4y)}}{12\,D_3} + \frac{i\sqrt{ax^2-b^2}\sqrt{a}\,by^2\,e^{(4y)}}{D_3}$$
$$- \frac{2i\sqrt{ax^2-b^2}\sqrt{a}\,\alpha^2b\,e^{(3y)}}{D_3} - \frac{5\,a\alpha x^2e^{(3y)}}{12\,D_3} + \frac{2i\sqrt{ax^2-b^2}\sqrt{a}\,\alpha by\,e^{(3y)}}{D_3} + \frac{i\sqrt{ax^2-b^2}\sqrt{a}\,\alpha^2b\,e^{(2y)}}{D_3} - \frac{\alpha b^2e^{(4y)}}{3\,D_3} + \frac{ax^2e^{(4y)}}{6\,D_3} + \frac{b^2ye^{(4y)}}{3\,D_3} + \frac{\alpha b^2e^{(3y)}}{3\,D_3} \tag{33}$$

$$\Psi_3 = \frac{\sqrt{ax^2-b^2}\sqrt{-\alpha e^y+ye^y+\alpha}\,\alpha^2e^{(\frac{7}{2}y)}}{4\,D_2} - \frac{\sqrt{ax^2-b^2}\sqrt{-\alpha e^y+ye^y+\alpha}\,\alpha ye^{(\frac{7}{2}y)}}{2\,D_2} + \frac{\sqrt{ax^2-b^2}\sqrt{-\alpha e^y+ye^y+\alpha}\,y^2e^{(\frac{7}{2}y)}}{4\,D_2}$$
$$- \frac{\sqrt{ax^2-b^2}\sqrt{-\alpha e^y+ye^y+\alpha}\,\alpha^2e^{(\frac{5}{2}y)}}{2\,D_2} + \frac{\sqrt{ax^2-b^2}\sqrt{-\alpha e^y+ye^y+\alpha}\,\alpha ye^{(\frac{5}{2}y)}}{2\,D_2} + \frac{\sqrt{ax^2-b^2}\sqrt{-\alpha e^y+ye^y+\alpha}\,\alpha^2e^{(\frac{3}{2}y)}}{4\,D_2}$$
$$+ \frac{i\sqrt{-\alpha e^y+ye^y+\alpha}\,\alpha b\,e^{(\frac{7}{2}y)}}{4\,D_2\sqrt{a}} - \frac{i\sqrt{-\alpha e^y+ye^y+\alpha}\,by\,e^{(\frac{7}{2}y)}}{4\,D_2\sqrt{a}} - \frac{i\sqrt{-\alpha e^y+ye^y+\alpha}\,\alpha b\,e^{(\frac{5}{2}y)}}{2\,D_2\sqrt{a}} + \frac{i\sqrt{-\alpha e^y+ye^y+\alpha}\,b\,e^{(\frac{7}{2}y)}}{4\,D_2\sqrt{a}} \tag{34}$$

$$\Psi_4 = \frac{\alpha e^{(2\,y)}}{4\,(\alpha^4xe^{(2\,y)} - 2\,\alpha^3xye^{(2\,y)} + \alpha^2xy^2e^{(2\,y)} - 2\,\alpha^4xe^y + 2\,\alpha^3xye^y + \alpha^4x - 2\,\alpha^3xe^{(2\,y)} + 4\,\alpha^2xye^{(2\,y)} - 2\,\alpha xy^2e^{(2\,y)} + 4\,\alpha^3xe^y - 4\,\alpha^2xye^y - 2\,\alpha^3x + \alpha^2xe^{(2\,y)} - 2\,\alpha xye^{(2\,y)} + xy^2e^{(2\,y)} - 2\,\alpha^2xe^y + 2\,\alpha xye^y + \alpha^2x)}$$
$$- \frac{e^{(3\,y)}}{4\,(\alpha^4xe^{(2\,y)} - 2\,\alpha^3xye^{(2\,y)} + \alpha^2xy^2e^{(2\,y)} - 2\,\alpha^4xe^y + 2\,\alpha^3xye^y + \alpha^4x - 2\,\alpha^3xe^{(2\,y)} + 4\,\alpha^2xye^{(2\,y)} - 2\,\alpha xy^2e^{(2\,y)} + 4\,\alpha^3xe^y - 4\,\alpha^2xye^y - 2\,\alpha^3x + \alpha^2xe^{(2\,y)} - 2\,\alpha xye^{(2\,y)} + xy^2e^{(2\,y)} - 2\,\alpha^2xe^y + 2\,\alpha xye^y + \alpha^2x)} \tag{35}$$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type "I"